EXPERIMENT NO. 9

Aim : Case Study on Data Exploration and Data Visualization Using Power BI

THEORY:

Case Study: Supermarket Sales Analysis

1. Introduction

Data Exploration is the process of examining data to understand its structure, quality, patterns and relationships.

Data Visualization represents data graphically using charts, graphs and dashboards. It helps users understand complex data easily.

Power BI is a business intelligence tool used to analyze data and create interactive visualizations and dashboards.

Problem Statement

A supermarket wants to analyze its sales data to understand:

- Which product category has the highest sales?
- How do sales change over time?
- What is the contribution of each category?
- Is there a relationship between quantity sold and sales?

Sample Dataset

Invoice	Category	City	Quantity	Sales (₹)	Profit (₹)
1001	Electronics	Mumbai	5	25,000	4,000
1002	Grocery	Pune	10	12,000	2,500
1003	Clothing	Mumbai	7	18,000	3,000
1004	Electronics	Nashik	3	15,000	2,000
1005	Grocery	Pune	8	10,000	2,200
1006	Clothing	Nashik	6	14,000	2,400

Data Types

- **Categorical:** Category, City
- **Numerical:** Quantity, Sales, Profit
- **Identifier:** Invoice

Data Exploration

The following activities are performed during data exploration:

- Check number of rows and columns.
- Identify data types.
- Check missing values.
- Check duplicate records.
- Identify incorrect/inconsistent values.
- Calculate minimum, maximum and average values.
- Identify important patterns and relationships.

Data Preparation

Before visualization:

1. Remove duplicate records.
2. Handle missing values.
3. Correct data types.

Data Visualization Using Power BI

Analysis	**Visualization**	**Purpose**
Sales by category	Bar Chart	Compare categories
Sales by city	Column Chart	Compare cities
Monthly sales	Line Chart	Identify trends
Quantity vs Sales	Scatter Plot	Identify relationship

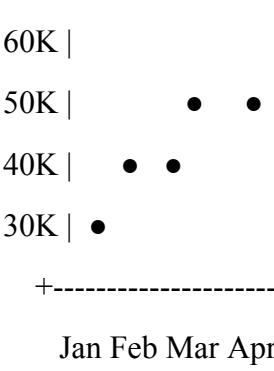
Example Charts

A. Sales by Category – Bar Chart



Interpretation: Electronics has the highest sales in this illustrative example.

B. Monthly Sales – Line Chart



Interpretation: The line chart helps identify increases and decreases in sales over time.

C. Quantity vs Sales – Scatter Plot

Purpose: To examine whether quantity sold and sales amount have a relationship.

Interpretation: A scatter plot helps identify whether the two numerical variables show a positive, negative or weak relationship.

9. Proposed Power BI Dashboard

SUPERMARKET SALES DASHBOARD			
TOTAL SALES **₹99,000**	**TOTAL PROFIT** **₹15,000**	**TOTAL QTY** **23 Units**	**NO. OF Records** **6**
SALES BY PRODUCT CATEGORY | | **Electronics | Grocery | Clothing**			
SALES BY CITY **Mumbai vs Pune**		**QUANTITY BY CATEGORY** **Electronics/Grocery/ Clothing**	
SALES vs PROFIT (Scatter Chart)			

Findings

From the sample illustrative analysis:

1. Electronics has the highest sales.
2. Sales can be compared effectively across different cities.
3. Monthly visualization helps identify sales trends.
4. Scatter plots help examine relationships between numerical attributes.

Advantages of Power BI

- Easy-to-understand visualizations
- Interactive dashboards
- Supports multiple chart types
- Helps identify trends and patterns
- Provides filtering and drill-down
- Useful for data analysis and decision-making

Conclusion

Data exploration and data visualization are important activities in the data mining process. Data exploration helps understand the structure, quality and characteristics of data, while visualization helps represent the discovered information in an understandable form. The proposed dashboard use visualizations to provide a summarized view of sales performance. Through such visualization, users can identify comparisons, trends, proportions and relationships in the dataset.